Institut d'Optique just outside of Paris. He was freshly returned from three years teaching in Cameroon, and he was looking for a research topic to earn his PhD while he worked as a lecturer at the institute. A professor mentioned an interesting seminar he had just heard given by an American physicist named Shimony, and from there Aspect found Bell's paper. "When I read the paper of Bell, I was absolutely fascinated. I thought it was the most exciting subject I had ever read," Aspect recalled. "It's like love at first sight.… So, at that point, I say, OK, I want to do my PhD on that." Aspect read Clauser and Freedman's paper and the conflicting results of Holt and Pipkin's experiment, and decided not to compete with them. "I was sure that somebody would settle the conflict much before I would start," Aspect said. "If I want to enter the game, I have to do something different. And I look carefully [at] the paper of Bell, and clearly in the conclusion, Bell said what was the important experiment to do. It was changing the orientation of the polarizers while the photons were in flight."

Bell's idea was simple in theory but enormously difficult to carry out in practice. When Clauser and the others had performed their tests of Bell's inequality, they had selected the polarizer angles randomly—but that random selection had happened *before* the pair of entangled photons was emitted from the source. In theory, the photons could have somehow detected those randomly selected settings before leaving the source, through some as yet unknown physics. If that had happened, there was no need to invoke nonlocality to explain the results of Clauser's experiment—some new purely local physics could explain it. The only way to rule this out would be to set the polarizers randomly while the entangled photons were already flying apart from each other. That way, no signal traveling at the speed of light could reach both photons after the polarizers had been set. "John Bell, I think, believed that…possible discrepancies with quantum predictions would show up in experiments where you rotated the [polarizers] rapidly," Clauser said later. The problem was that this required switching the polarizers enormously fast, faster than the time that it took light to travel from the photon source to the polarizers themselves. Typically, that distance was about ten meters, meaning that the polarizers needed to be switched in less than forty nanoseconds. The technical challenge was huge. "I began to think, carefully, how could I do it," Aspect recalled. "Finally I came to a conclusion that it might be possible." Aspect